

ANTWERP INTL, Belgium

WMO: 064500

Lat: 51.1903N

Lon: 4.4578E

Elev: 12

StdP: 101.18

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-4.1	-9.6	1.7	-4.9	-7.3	2.0	-3.1	12.2	9.1	11.0	9.6	2.2	70	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	29.8	20.6	27.6	19.5	25.6	18.6	21.4	27.9	20.3	26.1	19.4	24.4	3.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.9	24.4	18.2	13.1	23.2	17.3	12.4	22.0	62.1	28.0	58.4	26.3	55.1	24.5	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.5	8.3	7.3	DB	-9.0	33.6	2.8	1.7	-11.0	34.8	-12.6	35.8	-14.2	36.7	-16.2	37.9	(4)
			WB	-9.5	23.5	2.7	1.0	-11.4	24.2	-13.0	24.8	-14.5	25.4	-16.4	26.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	3.9	4.6	7.1	10.3	14.0	16.9	19.0	18.7	15.6	11.8	7.5	4.6
	DBStd	6.39	4.28	3.90	3.37	3.45	3.43	3.05	3.02	2.74	2.61	3.26	3.40	4.14
	HDD10.0	762	189	153	99	37	4	0	0	0	0	19	87	172
	HDD18.3	2721	446	384	347	240	140	63	27	28	90	204	324	427
	CDD10.0	1207	2	2	10	47	129	208	280	269	167	74	13	4
	CDD18.3	123	0	0	0	1	7	21	49	39	7	0	0	0
	CDH23.3	1024	0	0	0	12	77	187	390	308	48	3	0	0
	CDH26.7	280	0	0	0	1	12	48	117	95	7	0	0	0
Wind	WSAvg	3.4	4.0	3.9	3.8	3.3	3.3	3.2	3.1	2.9	3.0	3.3	3.4	3.8
Precipitation	PrecAvg	817	68	60	50	44	59	75	80	83	67	67	78	84
	PrecMax	1001	141	126	110	94	108	132	150	232	192	147	149	191
	PrecMin	629	4	14	4	0	16	25	25	9	11	21	8	26
	PrecStd	100	34	28	28	25	25	30	36	55	46	26	30	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	15.1	19.0	25.2	28.3	31.2	32.9	32.5	27.6	23.2	17.2	14.1
		MCWB	11.2	10.0	12.3	16.0	18.9	21.1	21.6	21.8	19.8	17.4	14.0	11.8
	2%	DB	12.0	12.7	16.3	21.8	25.5	27.8	29.8	29.3	24.5	20.0	15.1	12.8
		MCWB	10.2	9.8	11.4	14.4	16.9	19.7	20.7	20.4	18.4	16.0	12.7	11.0
	5%	DB	10.8	11.3	14.1	19.1	23.1	25.3	27.4	26.8	22.2	18.1	13.7	11.7
		MCWB	9.0	9.1	10.3	13.1	16.1	18.4	19.6	19.5	17.1	14.8	11.7	10.0
	10%	DB	9.5	10.1	12.4	16.6	20.7	23.1	25.3	24.3	20.4	16.6	12.5	10.2
		MCWB	7.9	8.2	9.5	11.8	15.0	17.2	18.7	18.3	16.0	14.0	10.7	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.4	13.4	17.2	19.9	22.2	22.7	22.9	20.7	17.7	14.4	12.3
		MCDB	12.8	13.5	17.2	23.2	26.3	29.2	30.6	29.8	25.8	21.7	16.2	13.4
	2%	WB	10.4	10.2	12.1	14.8	18.2	20.4	21.3	21.2	19.1	16.3	13.1	11.3
		MCDB	11.8	12.0	15.1	20.6	23.3	26.2	28.1	27.4	23.5	19.3	14.7	12.6
	5%	WB	9.2	9.4	10.9	13.4	16.6	19.1	20.3	20.0	17.7	15.3	12.0	10.1
		MCDB	10.6	11.1	13.6	18.1	21.5	24.4	26.3	25.5	21.0	17.6	13.6	11.6
	10%	WB	8.0	8.3	9.9	12.2	15.4	17.8	19.3	18.9	16.7	14.4	10.8	8.9
		MCDB	9.4	10.0	12.0	16.2	20.0	22.1	24.1	23.3	19.7	16.3	12.3	10.2
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.7	7.3	9.0	9.0	8.8	8.7	8.9	8.1	7.2	5.5	4.6
		MCDBR	5.4	7.5	10.7	13.7	13.4	13.0	12.9	13.5	11.4	9.4	6.5	5.1
		MCWBR	4.7	5.4	6.8	7.6	6.5	6.5	5.7	6.2	6.1	5.8	5.0	4.4
	5% WB	MCDBR	5.2	6.2	9.3	12.0	11.5	11.9	11.6	12.1	10.2	8.6	6.1	4.8
		MCWBR	4.8	4.9	6.2	7.0	6.3	6.5	5.6	6.1	5.9	5.6	5.0	4.3
Clear-Sky Solar Irradiance	taub	0.334	0.356	0.390	0.423	0.433	0.429	0.434	0.424	0.397	0.370	0.346	0.331	
	taud	2.369	2.336	2.242	2.180	2.199	2.231	2.220	2.267	2.331	2.393	2.377	2.340	
	Ebn at Noon	675	759	797	813	824	830	817	805	784	734	650	608	
	Edh at Noon	64	85	112	133	136	133	133	121	102	80	63	57	
All-Sky Solar Radiation	RadAvg	0.76	1.44	2.60	4.17	4.98	5.40	5.08	4.34	3.23	1.90	0.89	0.59	
	RadStd	0.10	0.28	0.37	0.46	0.44	0.45	0.60	0.41	0.31	0.15	0.09	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page